Chapter 5. Association and Independence of Two Variables

Pearson and Spearman correlation, Kendall's tau, permutation tests of independence

Contents

5.1 Correlation and simple linear regression **2**
- 5.1.1 Pearson's correlation coefficient . . . 2
- 5.1.2 Correlation and simple linear regression . . . 3
- 5.1.3 Spearman's rank correlation . . . 3

5.2 Tests based on correlation coefficients **4**
- 5.2.1 When normality can be assumed . . . 5
- 5.2.2 Permutation tests . . . 6
- 5.2.3 Permutation tests with Spearman's rank correlation . . . 7

5.3 Kendall's tau **8**
- 5.3.1 Tests of independence with Kendall's tau . . . 10

5.4 Linear association **11**

Overview

Inference about the relation between two variables is a very important problem in statistics. E.g. to see whether university entrance exam scores are related to academic performance at university, we infer the association between exam scores and university GPA. When both variables are numerical, the **correlation coefficient** is the standard measure of their relation. This chapter covers nonparametric correlation coefficients that can replace the sample correlation, and (distribution-free) tests of independence of two variables based on them.

Example 5.1

The data below are the teacher's ranking and the reading-test scores of 10 children. We want to know whether the teacher's evaluation agrees with the test scores. Is the sample correlation a good measure of association for these data?

teacher's rank	1	2	3	4	5	6	7	8	9	10
test score	10	20	25	50	80	82	85	90	95	100

Example 5.2

Figure 5.1 of the book is a scatter plot of the happiness score (Score) and the GDP per capita of the 13 happiest countries of the 2020 World Happiness Report. Does happiness increase with GDP?

5.1 Correlation and simple linear regression

5.1.1 Pearson's correlation coefficient

The most standard measure of association between random variables X, Y is Pearson's correlation coefficient. With $\mu_X = E(X)$, $\sigma_X^2 = \mathrm{Var}(X)$, $\mu_Y = E(Y)$, $\sigma_Y^2 = \mathrm{Var}(Y)$,

$$\rho = \mathrm{Corr}(X, Y) = \frac{E\{(X - \mu_X)(Y - \mu_Y)\}}{\sigma_X \sigma_Y} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

measures the *linear* association of the two variables, and satisfies:

- $-1 \le \rho \le 1$;
- if X, Y are independent, $\rho = 0$;
- $\rho > 0$ means positive correlation (Y tends to increase with X), $\rho < 0$ negative correlation.

Pearson's ρ is determined by the population of the pair (X, Y) observed together (here the joint distribution of (X, Y)). If the population is bivariate normal,[1] $\rho = 0$ means the two variables are independent. If not, even a small ρ does not mean independence. E.g. if $Y = (X - \mu_X)^2$, Y is exactly determined by X (a very strong association), but the correlation, which measures only linear association, can be $\rho = 0$.

From bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ the **sample correlation coefficient** is

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}.$$

[1] If a random vector $\mathbf{x} = (X_1, \ldots, X_p)'$ is multivariate normal with mean $\mu = (E(X_1), \ldots, E(X_p))'$ and covariance matrix $\Sigma = (\sigma_{ij})_{i,j=1,\ldots,p}$, $\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$, its joint density is $f(x_1, \ldots, x_p) = \frac{1}{\sqrt{|2\pi\Sigma|}} \exp\{-\frac{1}{2}(\mathbf{x} - \mu)'\Sigma^{-1}(\mathbf{x} - \mu)\}$.

Here $S_{XY} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$; divided by $n-1$ it is the sample covariance $\widehat{\mathrm{Cov}}(X,Y) = S_{XY}/(n-1)$, and S_{XX}, S_{YY} divided by $n-1$ are the sample variances. If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$ and the population has Pearson correlation ρ, the sample correlation r estimates ρ.

Whether or not the population is normal, Pearson's ρ or the sample r is the most frequently used measure of association, mainly because it measures only linear association and so is easy and intuitive to interpret. "Correlation coefficient" refers to Pearson's.

5.1.2 Correlation and simple linear regression

To study the association of two variables, the relation of the pairs (X_i, Y_i) can be modeled by simple linear regression:

$$Y_i = \alpha + \beta X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, n. \tag{5.1}$$

The least squares estimator $\hat{\beta}$ of the slope of the line $y = \alpha + \beta x$ is closely related to the sample correlation r:

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}} = r\sqrt{\frac{S_{YY}}{S_{XX}}}.$$

If the error ϵ_i in (5.1) is normal and X_i is normal too, (X_i, Y_i) is bivariate normal. Under this normality, the slope β and the correlation ρ are related by $\beta = \rho\sigma_X/\sigma_Y$ (more precisely $\beta = \mathrm{Cov}(X,Y)/\sigma_X^2 = \rho\sigma_Y/\sigma_X$).

Drawing the fitted simple regression line on the scatter plot as a trend line, its slope is a good measure of correlation.

Example 5.3 (Example 5.2 continued)

Adding the fitted simple regression line to the data on happiness score (Score) and GDP per capita gives an increasing trend line, so there seems to be a positive correlation.

5.1.3 Spearman's rank correlation

Computing the sample correlation after rank-transforming the data gives **Spearman's rank correlation coefficient** r_s. Rank-transform the variable $(X_1, \ldots, X_n)$ to $(R_1, \ldots, R_n)$ and $(Y_1, \ldots, Y_n)$ to $(S_1, \ldots, S_n)$, and form the rank-transformed data

$$\{(R_1, S_1), \ldots, (R_n, S_n)\};$$

their sample correlation, Spearman's rank correlation, is

$$r_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}.$$

It lies in $-1 \le r_s \le 1$; $r_s > 0$ means positive correlation and $r_s < 0$ negative. While the sample (Pearson) correlation measures only linear trends, Spearman's rank correlation measures any monotone increasing or decreasing trend, even nonlinear.

Example 5.4

To measure the association of "teacher's rank" and test score in Example 5.1, the sample correlation $r = 0.95$ shows a strong linear association. Spearman's rank correlation is $r_s = 1$: a perfect positive correlation (though not linear). The scatter plot (Figure 5.2 of the book) shows the relation clearly. Both are computed with R's `cor()`: `method = "pearson"` (or no argument) gives the sample correlation and `method = "spearman"` the rank correlation.

```
> ranks = 1:10
> score = c(10,20,25,50,80,82,85,90,95,100)
> cor(ranks,score)
[1] 0.948262
> cor(ranks,score, method = "spearman")
[1] 1
```

The sample correlation of the rank-transformed `ranks` and `score` is the rank correlation (`ranks` are already ranks):

```
> r = ranks
> (s = rank(score))
 [1]  1  2  3  4  5  6  7  8  9 10
> cor(r,s)
[1] 1
```

Spearman's rank correlation is a statistic computed from the data; there is no population parameter it corresponds to. If the two variables have a monotone trend (increasing or decreasing), $|r_s|$ is large; if they are independent or have no monotone trend, $r_s \approx 0$ tends to hold. Unlike the sample correlation, which applies only to numerical variables, it applies to ordinal categorical variables (e.g. very satisfied, satisfied, neutral, dissatisfied, very dissatisfied). Finally, the rank correlation is robust to outliers or influential points: in the scatter plot of Example 5.2, removing the last point on the right reveals a stronger monotone increase; if it is an outlier, the rank correlation is less affected by it.

Example 5.5

For the data of Example 5.2, the sample correlation is $r = 0.30$ and the rank correlation $r_s = 0.65$. The second variable `GDP` has ties; we check how they are handled.

```
> wh = read.csv("data/whappiness.csv")
> (r = rank(wh$Score))
 [1] 13 12 11 10  9  8  7  6  5  4  3  2  1
> (s = rank(wh$GDP.per.capita))
 [1]  8.5 11.0  8.5 12.0 10.0  7.0  2.0  6.0 13.0  4.0  5.0  3.0
[13]  1.0
> cor(r,s, method = "pearson")
[1] 0.6492441
> cor(wh$Score,wh$GDP.per.capita, method = "spearman")
[1] 0.6492441
```

5.2 Tests based on correlation coefficients

Consider tests of association of two variables using the sample correlation r or Spearman's r_s. The broadest hypotheses are

$$H_0 : \text{the two variables are independent} \quad \text{vs} \quad H_1 : \text{the two variables are related.}$$

If the population is bivariate normal, independence means $\rho = 0$, so the more specific hypotheses about ρ,

$$H_0 : \rho = 0 \quad \text{vs} \quad H_1 : \rho \neq 0, \tag{5.2}$$

are hypotheses of independence. One-sided alternatives $H_1 : \rho < 0$ or $H_1 : \rho > 0$ are also possible. Even if the population is not bivariate normal, (5.2) about ρ can be tested with the sample correlation.

5.2.1 When normality can be assumed

First the test based on the sample correlation r. When H_0 is true and normality holds, the increasing function of r

$$T = r\sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2} \tag{5.3}$$

is known to have exactly the t_{n-2} null distribution.

Proof (supplied). Condition on $X_1, \ldots, X_n$. Under bivariate normality with $\rho = 0$, $Y_1, \ldots, Y_n$ are i.i.d. normal and independent of the X's, i.e. the regression model (5.1) holds with normal errors and $\beta = 0$. In that model the usual t-statistic for $H_0 : \beta = 0$, $\hat{\beta}/\widehat{\mathrm{se}}(\hat{\beta})$, is t_{n-2}-distributed. Since $\hat{\beta} = S_{XY}/S_{XX}$, the residual sum of squares is $SSE = S_{YY} - S_{XY}^2/S_{XX} = S_{YY}(1-r^2)$ and $\widehat{\mathrm{se}}(\hat{\beta})^2 = SSE/((n-2)S_{XX})$, so

$$\frac{\hat{\beta}}{\widehat{\mathrm{se}}(\hat{\beta})} = \frac{r\sqrt{S_{YY}/S_{XX}}}{\sqrt{S_{YY}(1-r^2)/((n-2)S_{XX})}} = r\sqrt{\frac{n-2}{1-r^2}} = T.$$

The conditional distribution t_{n-2} does not depend on the X's, so it is also the unconditional distribution. ■

So if the observed sample correlation is r_{obs}, the two-sided p-value for (5.2) is

$$p = P(|R| \geq |r_{\mathrm{obs}}| \mid H_0) = P(|T| \geq |t_{\mathrm{obs}}|) = 2\{1 - F_{t_{n-2}}(|t_{\mathrm{obs}}|)\}, \tag{5.4}$$

where $F_{t_{n-2}}$ is the distribution function of t_{n-2} and $t_{\mathrm{obs}} = r_{\mathrm{obs}}\sqrt{\frac{n-2}{1-r_{\mathrm{obs}}^2}}$.

Even if the population is not normal, for large n the test is approximately exact by the CLT. In this situation, however, the t-test is a test about the correlation ρ, not a test of independence of the two variables.

Example 5.6

Treat the teacher's ranks of Example 5.1 as if they were continuous observations, and test the hypotheses (5.2). In words, H_0 is "the teacher's rank and the test score are unrelated" and H_1 is "they are related." First compute the sample correlation r_{obs}:

```
> n = 10
> (r.obs = cor(ranks, score))
[1] 0.948262
```

Using (5.4), the value of the t-statistic and the two-sided p-value:

```
> (t = r.obs * sqrt((n-2)/(1-r.obs^2)))
[1] 8.447828
> 2*(1-pt(t, df = n-2))
[1] 2.944375e-05
```

A very small p-value: strong evidence that the teacher's rank and the test score are related. (But here the teacher's rank is treated as continuous, so the normality the t-test requires is not satisfied and this is not a good test.) `cor.test()` gives the same result:

```
> cor.test(ranks,score)

        Pearson's product-moment correlation

data:  ranks and score
t = 8.4478, df = 8, p-value = 2.944e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7907589 0.9880014
```

```
sample estimates:
     cor
0.948262
```

The confidence interval in this output is an asymptotically approximate interval derived from (5.3).

5.2.2 Permutation tests

When the sample is not large enough for the CLT approximation, a permutation test with r as statistic can be performed. Under H_0: "the two variables are independent," the joint distributions of (X_1, Y_1) and (X_1, Y_2) are the same, so shuffling only one of the variables (e.g. Y) gives exchangeable observations. (If the variables are not independent, (X_1, Y_1) and (X_1, Y_2) have different distributions.) So for a permutation test of correlation, keep the "X" observations fixed, shuffle all n "Y" observations, and use the permutation distribution of the sample correlations r_i of the shuffles as the null distribution of r.

The table below shows, for the data of Example 5.1, two of the $n!$ possible shuffles and the statistics computed from them:

child	1	2	3	4	5	6	7	8	9	10	correlation
teacher's rank (x)	1	2	3	4	5	6	7	8	9	10	
test score (y)	10	20	25	50	80	82	85	90	95	100	$r_{\text{obs}} = 0.95$
shuffled scores	20	10	25	100	90	80	95	50	85	82	$r_1 = 0.64$
shuffled scores	80	10	100	85	90	25	50	82	20	95	$r_2 = -0.02$

The permutation test with r compares H_0: "the two variables are independent" with $H_1 : \rho \neq 0$ (or one-sided). The null is not "$\rho = 0$": if $\rho = 0$ but the variables are dependent, the shuffles above are not exchangeable. The alternative is stated specifically as "linear association" because r measures only linear correlation, so for nonlinear but related variables the power may be very low.

Example 5.7

With the data of Example 5.1, test independence of the two variables by the permutation test with the sample correlation r. All possible shuffles number $10! = 3628800$, too many, so we run a random permutation test. The R code repeats $M = 1999$ random permutations, shuffling only the values of `score` each time and recomputing the sample correlation under the null.

```
> M = 1999
> r.perm = replicate(M, {
+   perm.id = sample(1:n,n,replace = FALSE)
+   cor(ranks, score[perm.id])})
> (p.perm = 2*mean(c(r.obs,r.perm) >= r.obs))
[1] 0.001
```

(The book's figure shows the random permutation null distribution with the observed statistic `r.obs`.)

For large n the permutation distribution of r can be approximated by a normal. In the permutation distribution of

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{S_{XX} S_{YY}}},$$

the X_i are fixed and only the $(Y_1, \ldots, Y_n)$ are redrawn by random sampling without replacement from the finite population $\{Y_1, \ldots, Y_n\}$. Under any random draw (any shuffle) S_{XX} and S_{YY} do not change. The population mean of this finite population is $\bar{Y}$, so $E(Y_i) - \bar{Y} = 0$ and

$$E(r) = \frac{1}{\sqrt{S_{XX} S_{YY}}} \sum_{i=1}^{n} (X_i - \bar{X})\{E(Y_i) - \bar{Y}\} = 0.$$

Moreover, whatever the actual observations (X_i, Y_i), the variance of the permutation distribution is always

$$\mathrm{Var}(r) = \frac{1}{n-1}. \tag{5.5}$$

Proof of (5.5) (supplied). Let $a_i = X_i - \bar{X}$, $b_i = Y_i - \bar{Y}$ (so $\sum a_i = \sum b_i = 0$) and let π be a uniformly random permutation; $r = \sum_i a_i b_{\pi(i)} / \sqrt{S_{XX} S_{YY}}$. For each i, $b_{\pi(i)}$ is uniform on $\{b_1, \ldots, b_n\}$, so $\mathrm{Var}(b_{\pi(i)}) = \frac{1}{n} \sum b_k^2 = S_{YY}/n$; for $i \neq j$, $(b_{\pi(i)}, b_{\pi(j)})$ is uniform over ordered pairs of distinct indices, so $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = \frac{1}{n(n-1)} \sum_{k \neq l} b_k b_l = -\frac{S_{YY}}{n(n-1)}$. Using $\sum_{i \neq j} a_i a_j = -\sum_i a_i^2 = -S_{XX}$,

$$\mathrm{Var}\Big(\sum_i a_i b_{\pi(i)} \Big) = S_{XX} \frac{S_{YY}}{n} + S_{XX} \frac{S_{YY}}{n(n-1)} = \frac{S_{XX} S_{YY}}{n-1}.$$

Dividing by $S_{XX} S_{YY}$ gives (5.5). ■

These mean and variance are of the permutation distribution of r conditional on the observations (X_i, Y_i) (random shuffles of the order of the Y_i), not of the sampling distribution of r (with random (X_i, Y_i)).

For r following its permutation distribution, the finite-population CLT gives for large n

$$r\sqrt{n-1} \xrightarrow{d} N(0,1) :$$

the permutation distribution of r is approximately $N(0, \frac{1}{n-1})$, which gives a test. Figure 5.3 of the book compares the permutation distribution of Example 5.7 ($n = 10$) with the density of $N(0, 1/9)$: even for this very small sample the difference is hard to see by eye (in fact it is larger in the tails). With more observations the approximation is better.

5.2.3 Permutation tests with Spearman's rank correlation

Spearman's rank correlation is the sample correlation of the rank-transformed (R_i, S_i) of (X_i, Y_i), so tests proceed exactly like the permutation test with the sample correlation above. (Even the mean and variance of the large-sample approximation are the same.) The hypotheses here are

$$\begin{aligned} &H_0 : \text{the two variables are independent} \quad \text{vs} \\ &H_1 : \text{the two variables have a monotone increasing or decreasing correlation.} \end{aligned} \tag{5.6}$$

Correlation measures only linear association, while Spearman's rank correlation measures monotone relations, even nonlinear. A test with the rank correlation and its null distribution is unchanged by any monotone transformation of the observations (X_i, Y_i) (e.g. $x_i \mapsto x_i^2$, $y_i \mapsto \sqrt{y_i}$ when all x_i, y_i are positive), so it uses a distribution-free statistic and is a distribution-free test.

Example 5.8

For the data of Example 5.2, investigate with a permutation test based on the rank correlation whether the two variables are related. The null distribution is approximated by $M = 1999$ random permutations. The R code rank-transforms in advance, saving the work of ranking in every repetition.

```
> n = nrow(wh)
> r = rank(wh$Score)
> s = rank(wh$GDP.per.capita)
> (rs.obs = cor(r,s))
[1] 0.6492441
> rs.perm = replicate(1999, cor(r, s[sample(1:n,n,replace = F)])
  )
> (p.perm = 2*mean(c(rs.obs,rs.perm) >= rs.obs))
[1] 0.02
```

The p-value is 0.02, so at the 5% level we conclude the two variables are not independent. One may also use

```
cor.test(r,s, method = "spearman")
```

(these data have ties, so `cor.test()` cannot compute an exact p-value).

5.3 Kendall's tau

Another distribution-free measure of association of two variables is **Kendall's tau** (τ).

For bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ Kendall's tau is defined as follows. If two variables are positively correlated, then picking two observations (X_i, Y_i) and (X_j, Y_j), one point tends to be lower left and the other upper right: as one variable increases so does the other, and the pair is **concordant**. If they are negatively correlated, for two observations one point tends to be upper left and the other lower right: as one increases the other decreases, and the pair is **discordant**. In the data of Figure 5.2 all pairs of observations are concordant.

The average over all pairs of

$$U_{ij} = \begin{cases} +1, & (X_i - X_j)(Y_i - Y_j) > 0; \\ -1, & (X_i - X_j)(Y_i - Y_j) < 0; \\ 0, & (X_i - X_j)(Y_i - Y_j) = 0, \end{cases} \qquad 1 \le i < j \le n, \tag{5.7}$$

which is $+1$ for concordant and -1 for discordant pairs, is Kendall's tau computed from the sample,

$$\tau_n = \frac{1}{\binom{n}{2}} \sum_{i<j} U_{ij}. \tag{5.8}$$

With N_{Conc} concordant and N_{Disc} discordant pairs, $\sum_{i<j} U_{ij} = N_{\text{Conc}} - N_{\text{Disc}}$. If there are no "tied" pairs, i.e. none with $(X_i - X_j)(Y_i - Y_j) = 0$, then $N_{\text{Conc}} + N_{\text{Disc}} = \binom{n}{2}$, so

$$\tau_n = 2\frac{N_{\text{Conc}}}{\binom{n}{2}} - 1.$$

Example 5.9

Compute Kendall's τ_n for the data of Example 5.2. First generate the ids of the $\binom{13}{2} = 78$ pairs of observations, then compute the U_{ij}:

```
> n = nrow(wh)
> allpairs = gtools::combinations(n,2,1:n)
> head(allpairs,3)
     [,1] [,2]
[1,]    1    2
```

```
[2,]    1    3
[3,]    1    4
> U = apply(allpairs,1,function(pair.id){
+   sign( diff(wh$Score[pair.id]) * diff(wh$GDP.per.capita[pair.
    id]) )
+ })
```

There are $\#\{U_{ij} = 1\} = 58$ concordant pairs, $\#\{U_{ij} = -1\} = 19$ discordant pairs, and $\#\{U_{ij} = 0\} = 1$ tied pair. By (5.8):

```
> (tau = sum(U) / choose(n,2))
[1] 0.5
```

R's `cor()` with `method = "kendall"` gives a slightly different value. With ties, `cor()` computes Kendall's tau by a method of Kendall (1945) slightly different from (5.8); without ties the results agree.

```
> cor(wh$Score,wh$GDP.per.capita, method = "kendall")
[1] 0.5032363
```

Kendall's τ_n resembles the rank correlation r_s but differs. In common, both measure the monotone relation of two variables (on the same $[-1, 1]$ scale as Pearson's correlation) and are robust to outliers. The rank correlation is the sample correlation of the rank-transformed data, while Kendall's statistic uses sign-transformed data, so the values necessarily differ. The most important difference is that Kendall's tau has a corresponding parameter (the rank correlation does not). For two independent random vectors $(X, Y), (X', Y')$ drawn from a population, Kendall's tau parameter of the population is

$$\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\} = 2P\{(X - X')(Y - Y') > 0\} - 1$$

(the second form for continuous populations). The parameter τ is interpreted as:

- $\tau > 0$: the proportion (probability) of concordant pairs exceeds that of discordant pairs, a positive correlation;
- $\tau < 0$: a negative correlation;
- $\tau = 0$: concordant and discordant pairs are equally likely, and the two variables are unrelated (in this sense).

The parameter τ and the statistic τ_n play a role like correlation but are generally not called correlation coefficients, only "Kendall's tau"; as a nonparametric measure of correlation, being robust, it is more widely used than the rank correlation. If the population is bivariate normal, τ is determined by the population parameter ρ (correlation):

$$\tau = \frac{2}{\pi} \arcsin(\rho). \tag{5.9}$$

Proof of (5.9) (supplied). $(U, V) = (X - X', Y - Y')$ is bivariate normal with mean 0 and correlation ρ; standardize so $U = Z_1$, $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$ with Z_1, Z_2 i.i.d. $N(0, 1)$. Write $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$ with Θ uniform on $[0, 2\pi)$, and $\rho = \sin\phi$, $\sqrt{1 - \rho^2} = \cos\phi$, $\phi = \arcsin\rho \in [-\pi/2, \pi/2]$. Then $V = R\sin(\Theta + \phi)$, and $UV > 0$ iff $\cos\Theta$ and $\sin(\Theta + \phi)$ have the same sign. The set of such Θ is two arcs of total length $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\arcsin\rho}{\pi}$ (Sheppard's formula), and $\tau = 2P(UV > 0) - 1 = \frac{2}{\pi}\arcsin\rho$. ■

If the data are normal but contain outliers, rather than estimating ρ by the sample correlation r, it is more robust to estimate τ by Kendall's τ_n and estimate ρ by $\hat{\rho} := \sin(\frac{\pi}{2}\tau_n)$.

5.3.1 Tests of independence with Kendall's tau

The null hypothesis "the two variables are independent" of (5.6) against the alternative "the two variables have a monotone increasing or decreasing relation" can be tested with Kendall's τ_n. Fixing one variable and shuffling the other (as in the permutation test with correlation), generating the permutation distribution of τ_n and computing p-values gives an exact permutation test. When H_0 is true (the variables are independent), $\tau = 0$, so if the test fails to reject H_0 we may accept $\tau = 0$; if it accepts the alternative, there is strong evidence that $\tau \neq 0$.

Without a permutation test, one can use as null distribution the sampling distribution of τ_n when H_0 (where only $\tau = 0$ is needed) is true; this sampling distribution is in fact the permutation distribution above.[2] Since τ_n depends only on the signs of differences of observations,[3] its sampling distribution (permutation distribution) is distribution-free, but for large n the exact sampling distribution is hard to obtain. So with many observations, the traditional test of independence with Kendall's tau uses the large-sample normal approximation

$$\tau_n \dot{\sim} N\Big(0, \frac{4n+10}{9n(n-1)}\Big) \tag{5.10}$$

as null distribution.[4]

Variance in (5.10) (supplied). Assume independence and continuous margins. Each U_{ij} is ± 1 with probability $1/2$, so $EU_{ij} = 0$ and $\operatorname{Var} U_{ij} = 1$. Pairs $\{i,j\}, \{k,l\}$ with no common index give independent U's. For pairs sharing one index, e.g. U_{12}, U_{13}, independence of X and Y gives $E(U_{12}U_{13}) = E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\, E[\operatorname{sgn}(Y_1 - Y_2)\operatorname{sgn}(Y_1 - Y_3)]$. Conditional on X_1, $\operatorname{sgn}(X_1 - X_2)$ has mean $2F(X_1) - 1$, so the first factor is $E(2F(X_1) - 1)^2 = E(2V - 1)^2 = 1/3$ (V uniform), and likewise the second. So $\operatorname{Cov}(U_{12}, U_{13}) = 1/9$. There are $\binom{n}{2} \cdot 2(n-2)$ ordered pairs of pairs sharing exactly one index, so

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2} + \binom{n}{2}\frac{2(n-2)}{9} = \binom{n}{2}\frac{2n+5}{9},$$

and $\operatorname{Var}(\tau_n) = \frac{2n+5}{9\binom{n}{2}} = \frac{2(2n+5)}{9n(n-1)} = \frac{4n+10}{9n(n-1)}$. Asymptotic normality follows from the theory of U-statistics (Hoeffding). ■

Figure 5.4 of the book compares, for $n = 10$, the null distribution of Kendall's statistic (Kendall's tau distribution, equal to the permutation distribution) with the approximating normal (5.10). The normal approximation is fairly good even for small samples, but the tails differ slightly.

Example 5.10

To test independence of the two variables of Example 5.2 with Kendall's tau, we run (1) a permutation test, (2) the test with Kendall's tau distribution and (3) the test with the normal approximation, computing from the ranks of the observations only.

```
> n = nrow(wh)
> r = rank(wh$Score)
> s = rank(wh$GDP.per.capita)
> (tau.obs = cor(r,s, method = "k"))
[1] 0.5032363
```

These data have ties, so the permutation test is the most accurate, but there are too many shuffles, so we run a random permutation test with $M = 1999$ shuffles (computing the p-

[2] Without ties, the permutation distribution of τ_n is also called Kendall's tau distribution. R's `SuppDists` package gives its probability mass function via `dKendall()`.

[3] $(X_i - X_j)(Y_i - Y_j) > 0$ when the signs of $X_i - X_j$ and $Y_i - Y_j$ agree, so τ_n depends only on signs.

[4] For the computation of the variance of τ_n see Exercise 5.4 of the book; a derivation is supplied below.

value assuming the permutation distribution is symmetric). With T_n following Kendall's tau distribution, the p-value of the test with Kendall's tau distribution is $2P(T_n \geq \tau_n)$. With ties this p-value is not exact, but with only one tied pair the test is approximately exact. `pKendall()` of the `SuppDists` package gives the distribution function of Kendall's tau distribution. Finally the test with the normal approximation compares the observed τ_n with (5.10).

```
> # (1) random permutation
> tau.perm = replicate(1999, cor(r, s[sample(1:n,n,replace = F)
    ], method = "k"))
> p.perm = 2*mean(c(tau.obs,tau.perm) >= tau.obs)
>
> # (2) Kendall's tau distribution
> p.tau = 2*(1- SuppDists::pKendall(tau.obs,n))
>
> # (3) normal distribution
> p.z = 2*(1 - pnorm(tau,sd = sqrt(4*n +10)/sqrt(9*n*(n-1)) ))
>
> c(p.perm, p.tau, p.z)
[1] 0.02000000 0.01022864 0.01734336
```

Whichever test is used, at the 5% level H_0: "the two variables are independent" is rejected.

5.4 Linear association

The three measures of association above measure linear or monotone association, and tests of the presence of such association can be based on them. If the population is bivariate normal (Figure 5.5(a) of the book), only linear association exists between the variables, so the t-test for Pearson's correlation is the most appropriate and most efficient. Even when there is only linear correlation, if the marginal distributions have heavy tails (Figure 5.5(b)), tests with Spearman's rank correlation or Kendall's tau are more powerful than the t-test with the sample correlation. When the relation is not linear but monotone increasing (or decreasing) (Figure 5.5(c)), it is appropriate to quantify and test the trend with Spearman's rank correlation or Kendall's tau.

Example 5.11

The data of Figure 5.5 of the book were generated from $n = 100$ bivariate normal observations:

(a) (X, Y) bivariate normal with correlation $\rho = 0.9$;

(b) the data of (a) transformed to heavy tails, $(X, Y) \mapsto (X^3, Y^3)$ (the book prints X^4, Y^4; the plotted data take negative values and the rank statistics agree with (a), so the map is a monotone one such as the cube);

(c) the data of (a) transformed differently, with heavy tails, $(X, Y) \mapsto (\text{sgn}(X)X^2, -(Y - 4)^4)$;

(d) $X \sim N(0, 1)$, $Y = \sin(3X) + \epsilon$, $\epsilon \sim N(0, (0.2)^2)$.

For each data set we computed the sample correlation, the rank correlation and Kendall's statistic, and p-values of tests of independence with the large-sample approximation. Why are the rank correlation and Kendall's tau the same for data (a), (b), (c)? (All three maps are increasing in each coordinate on the range of the data, since $Y < 4$ there, so the ranks do not change.)

	sample correlation (p)	rank correlation (p)	Kendall's tau (p)
(a)	0.90 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(b)	0.82 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(c)	0.81 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(d)	0.005 ($p = 0.95$)	0.13 ($p = 0.2$)	0.09 ($p = 0.17$)

When the relation of the two variables is clear but not linear (Figure 5.5(d)), all three measures are near 0, so whichever statistic is used, the independence test fails to reject H_0: "the two variables are independent." Remember that here the failure to reject is not because H_0 is true but because all three tests lack power against nonlinear relations.